

Nova Electronics Corp — Marketing Value Creator (MVC)

System Architecture, C4 Component Models & Design Specifications

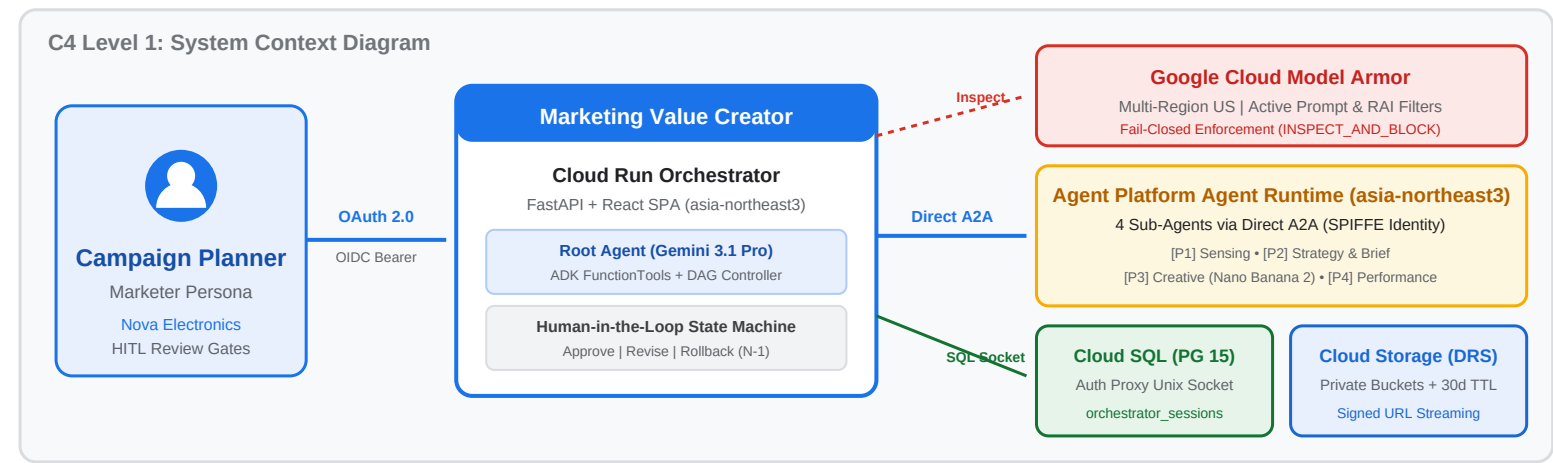
Forward Deployed Engineer: Ryan Ahn (ryanahn@) • Capstone Project • September 2026

Executive Architecture Summary: Marketing Value Creator (MVC) is an enterprise generative AI campaign planning and simulation platform designed to compress Nova Electronics Corp's manual 4-to-6-week multi-agency briefing workflow into an interactive, multi-agent simulation taking under 15 seconds of compute. The platform operates under strict brand and security guardrails with 100% mathematical budget conservation and zero arithmetic hallucination.

Architectural Dimension	Technical Design Implementation	Business & Operational Benefit
Topology & Deployment	FastAPI Orchestrator on Cloud Run (asia-northeast3) serving React 19 SPA static assets under a unified origin with Direct VPC Egress and Zero-Trust Firewall (Default Deny + Whitelist).	Zero CORS complexity, sub-50ms regional latency in Seoul, and automatic scale-to-zero cost efficiency.
Multi-Agent Orchestration	4 specialized sub-agents on Agent Platform Agent Runtime orchestrated via Direct A2A protocol (SPIFFE identity) with ADK Root Agent FunctionTools.	Modular domain separation ([P1] Sensing, [P2] Strategy, [P3] Creative, [P4] Performance) with zero context dilution.
Asymmetric Model Tiering	Gemini 3.5 Flash Lite for subagents (P1, P2, P4); Nano Banana 2 Lite (gemini-3.1-flash-lite-image) for P3 visuals; Gemini 3.1 Pro for root orchestration and evaluation.	FinOps optimization: \$0.0455 / run actual vs \$0.10 target ceiling (54.5% below target).
Security & Guardrails	Google Cloud Model Armor (version1-guardrails in multi-region US) with active prompt/RAI/SDP filters, fail-closed policy, Google OAuth 2.0 OIDC, and private Cloud SQL.	Zero public DB exposure, 100% prompt injection defense, and enterprise brand safety compliance.
Quality Flywheel & CI/CD	Cloud Build Automated Dual Gate (119 passing tests: 102 unit + 17 integration + pre-prod eval gate on 9 golden scenarios with LLM Judge >= 4.0/5.0).	Blocks release regressions automatically; enforces 100.0% budget conservation and 100% schema validity.

1. System Context Diagram (C4 Level 1)

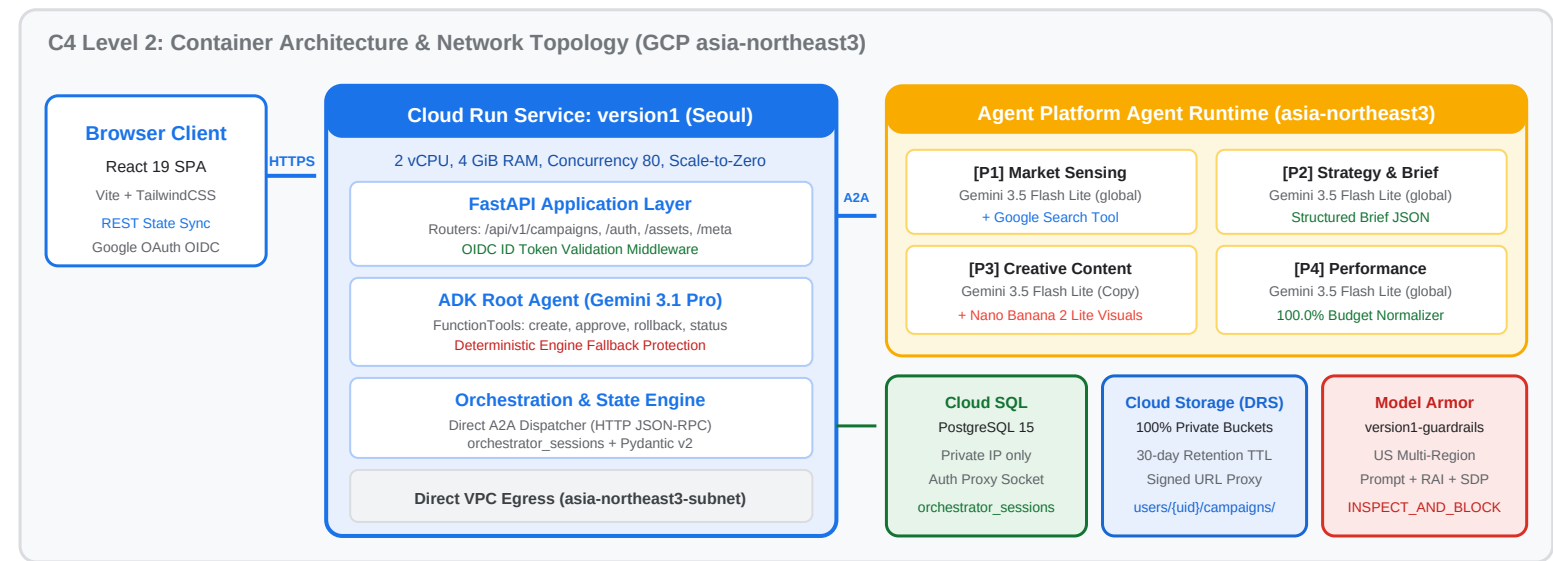
Maps the overarching boundary of the MVC system, user personas (Marketer / Campaign Planner), and external cloud dependencies.



Boundary Component	Protocol / Integration	Security & Trust Boundary	Data Classification & Retention
Campaign Planner (User)	HTTPS Browser / React 19 SPA	Google OAuth 2.0 OIDC Bearer Token verification on Cloud Run endpoint.	Marketer email and role (Viewer / Editor / Admin).
MVC Orchestrator	FastAPI on Cloud Run (asia-northeast3)	Direct VPC Egress; IAM Workload Identity; isolated container runtime.	Session state & intermediate stage deliverables (30-day TTL).
Agent Platform Agent Runtime	Direct A2A JSON-RPC Protocol	SPIFFE-based Agent Identity; authenticated internal RPC	Domain deliverables: market sensing, brief, creative, insights.
Google Cloud Model Armor	REST API (US Multi-Region)	Fail-closed policy (INSPECT_AND_BLOCK); pre-execution prompt & RAI inspection.	Transient inspection logs; security violation audit trails.

2. Container Architecture & Network Topology (C4 Level 2)

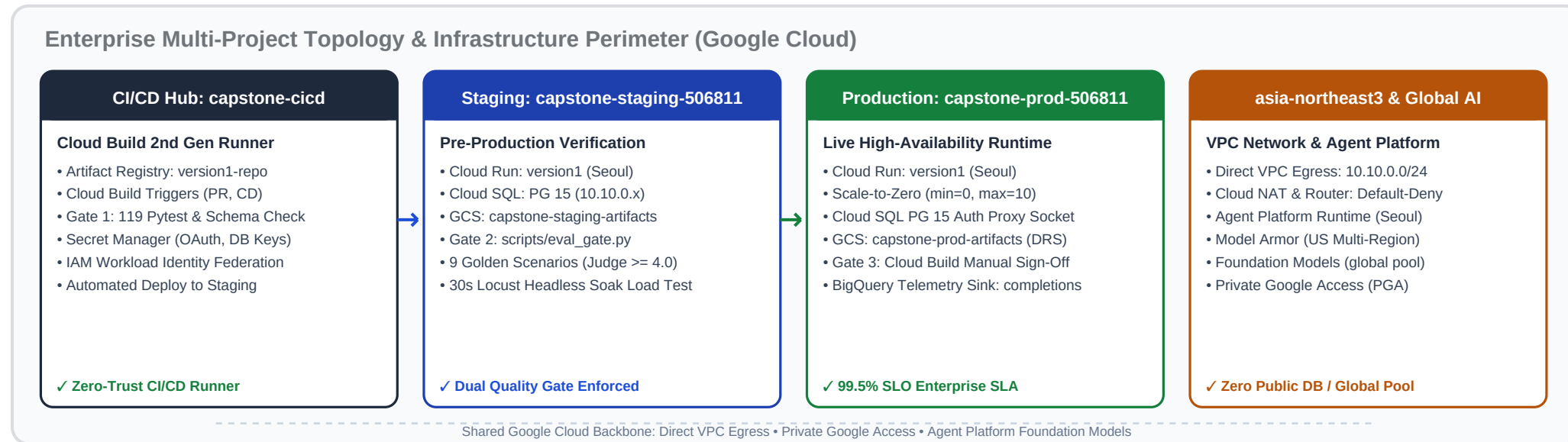
Detailed layout of Cloud Run, Direct VPC Egress, Cloud SQL Auth Proxy, and external Agent Platform model services.



Container / Resource	Sizing & Runtime Specs	Network & Security Controls	Design Rationale
Cloud Run Service	2 vCPU, 4 GiB RAM, concurrency 80, min 0, max 10	Direct VPC Egress (asia-northeast3-subnet 10.10.0.0/24); Zero-Trust Firewall (Default-Deny Ingress/Egress, Whitelist: 443, 3307, 53)	Eliminates VPC connector bottleneck; enables private IP database routing.
Cloud SQL PostgreSQL 15	db-f1-micro (shared vCPU, 0.6 GiB RAM)	Private IP only; Auth Proxy Unix domain socket at /cloudsql	Zero public IP attack surface; durable session persistence in orchestrator_sessions.
Agent Platform Foundation Models	Gemini 3.5 Flash Lite + Gemini 3.1 Pro + Nano Banana 2 Lite	Endpoints explicitly pinned to location='global'	Accesses worldwide GPU/TPU pool, avoiding regional model quota and 404 errors.

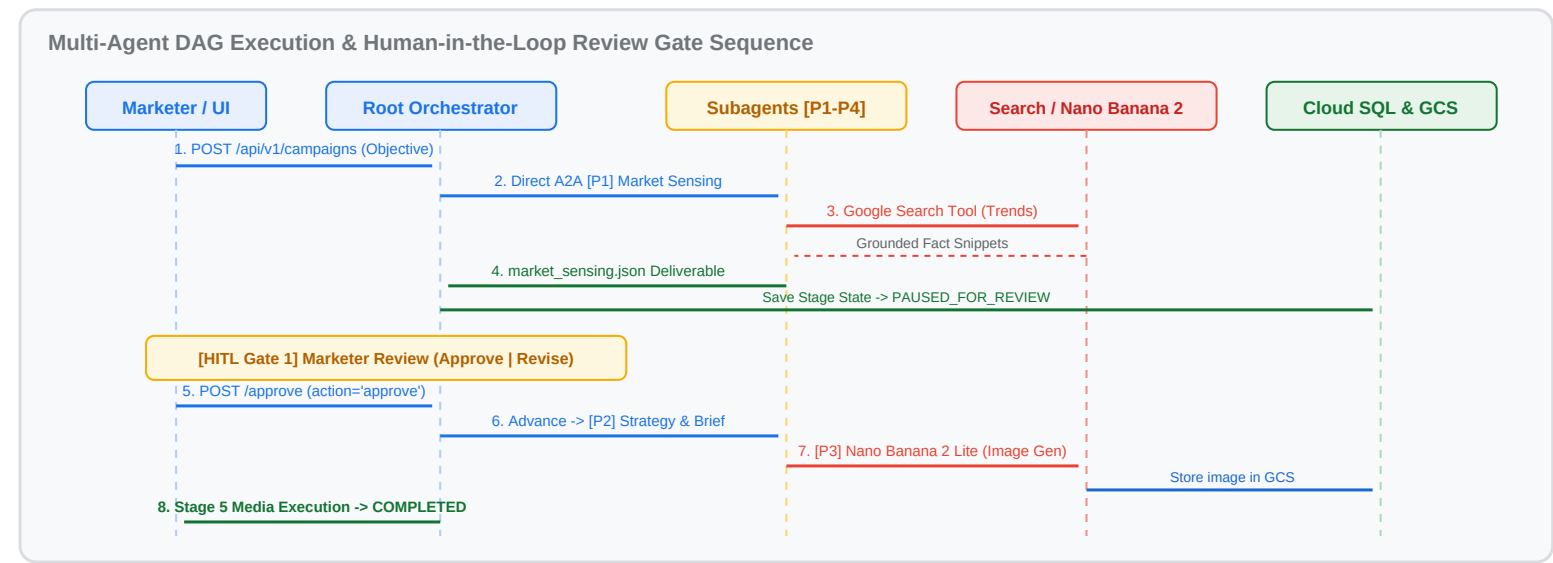
3. Multi-Project GCP Environment & Enterprise Infrastructure Architecture

Enterprise multi-project topology, Direct VPC Egress, Cloud SQL Auth Proxy, and Private DRS Artifact Storage.



4. Multi-Agent DAG & Human-in-the-Loop Review Sequence

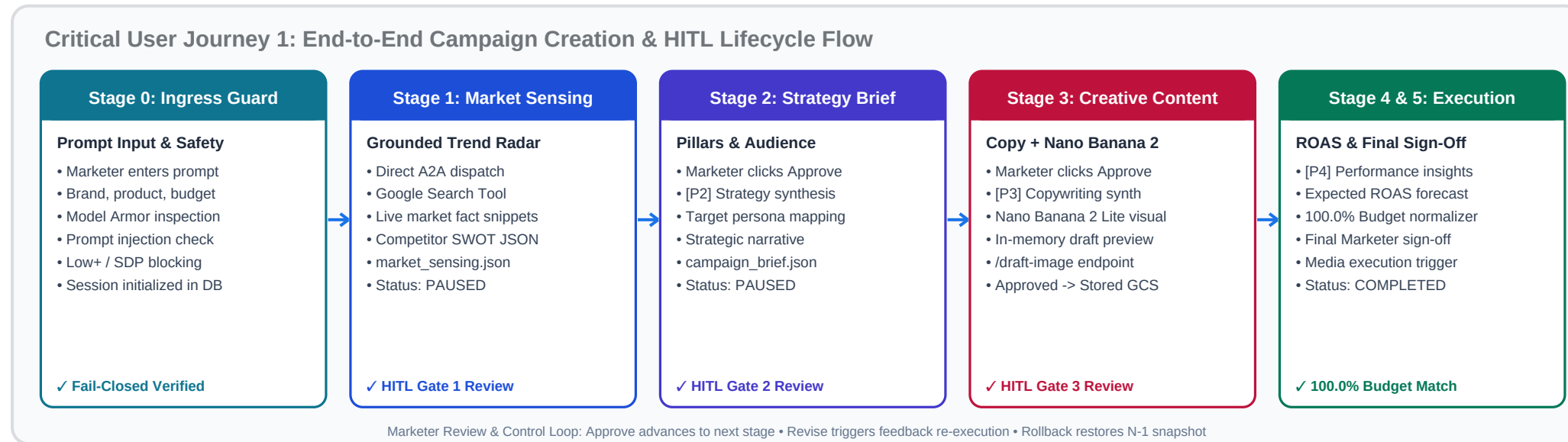
Illustrates sequential subagent execution across P1-P4 and the asynchronous Human-in-the-Loop review gates.



Execution Stage	Agent & Tool Assigned	Human Review Interaction (HITL)	Invariance / SLA Constraint
Stage 1: Market Sensing	Gemini 3.5 Flash Lite + Google Search Grounding Tool	Marketer inspects competitor findings and trend radar.	Turn latency < 3.0s; live grounded market facts.
Stage 2: Strategy & Brief	Gemini 3.5 Flash Lite (Structured Synthesis)	Marketer approves strategic narrative or provides revision feedback.	100% Pydantic validation on campaign_brief.json
Stage 3: Creative Content	Gemini 3.5 Flash Lite (Copy) + Nano Banana 2 Lite (gemini-3.1-flash-lite-image)	Marketer reviews copywriting and visual mockup image.	Visual generation latency < 8.0s; asset stored in DRS GCS.
Stage 4: Performance Insights	Gemini 3.5 Flash Lite + Deterministic Budget Normalizer	Marketer inspects ROAS projections and channel allocation.	100.0% Budget Conservation (Sum of channels == Total Budget).

5. Critical User Journey 1 (CUJ-1): End-to-End Campaign Creation & HITL Review

Step-by-step user and multi-agent interaction lifecycle from natural language objective to approved campaign.



Journey Phase	User Action & UI Event	Orchestrator & Subagent Action	Verification & Contract SLA
Phase 1: Inception & Pre-Flight	Enters campaign objective, budget (\$/KRW), and product in React 19 UI -> Clicks 'Create Campaign'.	Passes prompt to Model Armor; validates schema; creates session record in Cloud SQL orchestrator_sessions.	Prompt injection score < LOW; HTTP 400 rejection on violation; pre-flight latency < 450ms.
Phase 2: Sensing & Strategy	Reviews competitor matrix, SWOT findings, and trend radar -> Clicks 'Approve Stage'.	[P1] executes Google Search grounding -> market_sensing.json; [P2] synthesizes campaign_brief.json.	100% Pydantic validation; intermediate state persisted; status set to PAUSED_FOR_REVIEW.
Phase 3: Creative & Visual	Reviews advertising headline, body copy, and visual mockup preview image in browser.	[P3] generates ad copy; Nano Banana 2 Lite renders visual; served via /draft-image; approved to GCS.	Visual generation latency < 8.0s; asset stored in DRS private GCS; signed URL 307 redirect.
Phase 4: Insights & Final Sign-Off	Inspects ROAS projections, CPA forecast, and channel budget allocation table -> Clicks 'Final Approval'.	[P4] computes KPIs; applies deterministic budget normalizer; finalizes campaign to COMPLETED.	Strict 100.0% mathematical budget conservation (sum == budget); zero arithmetic hallucination.

6. Critical User Journey 2 (CUJ-2): Marketer Revision, Rollback (N-1) & Fallback Recovery

Human feedback loops, deterministic single-step rollback (N->N-1), and resilient subagent fallback chains.

Critical User Journey 2: Human Revision, Deterministic Rollback & Fault-Tolerant Recovery

Flow A: Marketer Feedback Revision

Iterative Prompt & Context Refinement

- Marketer enters feedback in Web UI (e.g. adjust audience)
- POST /api/v1/campaigns/{sessionId}/approve (action='revise')
- Intermediate deliverable snapshotted to session_history
- Current subagent re-executes with user revision feedback
- Stage 3: Atomically purges in-memory draft visual cache
- State resets to PAUSED_FOR_REVIEW for re-inspection

✓ Atomic In-Memory Draft Cache Invalidation

Flow B: Deterministic Rollback (N -> N-1)

Single-Step Historical State Recovery

- Marketer clicks 'Rollback to Previous Stage' in UI
- POST /api/v1/campaigns/{sessionId}/rollback
- Verifies stage boundary (N in [2, 5]); fetches N-1 snapshot
- Restores deliverables JSONB and decrements current_stage
- Zero upstream re-computation; instantaneous recovery (<100ms)
- Decrementes revision_count; maintains strict audit trail

✓ Zero Hallucination • Idempotent State Restoration

Flow C: Runtime Fallback & Error Resilience

Multi-Tier Model & Engine Fault Tolerance

- Model Quota / Timeout: HTTP 429 / 503 / network timeout
- Exponential Backoff with Jitter (initial 1.0s, max 3 retries)
- Model Fallback: Gemini 3.5 Flash Lite -> 3.1 Pro -> Template
- A2A Fallback: Direct Agent Runtime -> Engine fallback
- Emits span trace to Cloud Trace & BigQuery completions sink
- Zero user-facing 500 internal server errors guaranteed

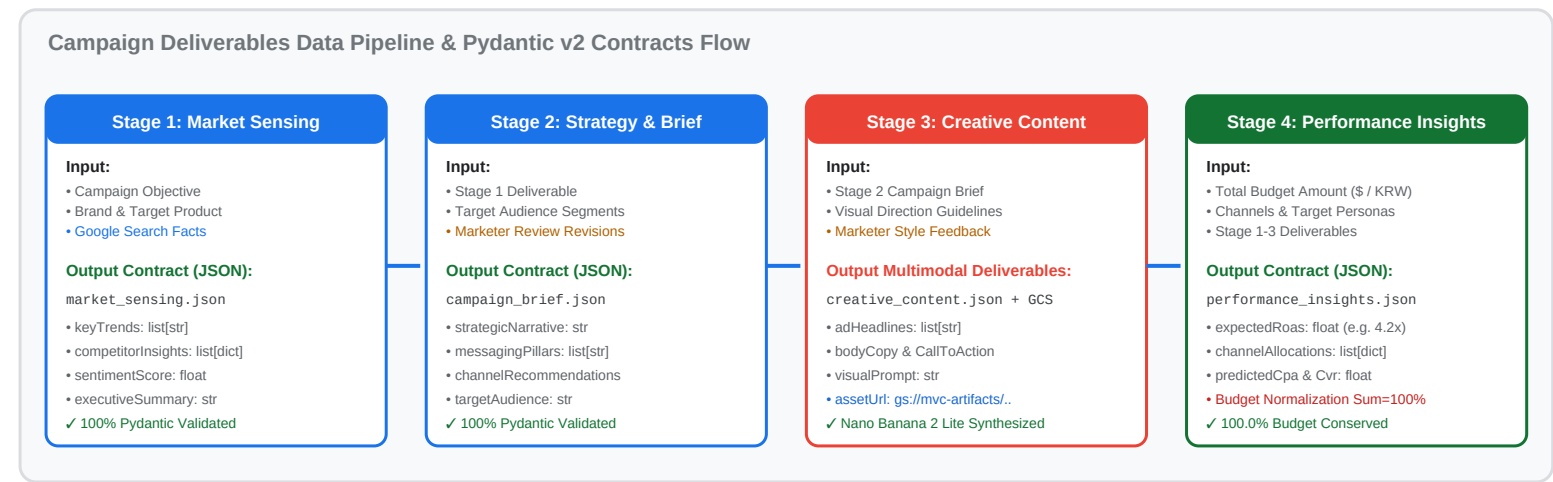
✓ 100% High-Availability Resilience SLA

Resilience Architecture: Zero State Loss • In-Memory Draft Protection • Automated Multi-Tier Graceful Degradation

Resilience Flow	Trigger Endpoint & Condition	State Machine Transition	Data Integrity & Fallback Guarantee
Marketer Feedback Revision	POST /api/v1/campaigns/{sessionId}/approve (action='revise', feedback='...')	Current stage subagent re-runs with feedback; revision counter increments; status resets to PAUSED_FOR_REVIEW.	Feedback persisted to session_history; in-memory draft image purged in Stage 3 to prevent stale visual reuse.
Deterministic N-1 Rollback	POST /api/v1/campaigns/{sessionId}/rollback (invoked from Stage N in [2, 5])	Current stage decrements N -> N-1; restores previous deliverable snapshot from session_history.	Instantaneous (<100ms); zero upstream re-computation; strictly idempotent rollback operation.
Model Fallback & Jitter Retry	Transient HTTP 429 / 503 / timeout on Agent Platform foundation model API	Transparent retry with jitter (1s, 2s, 4s); automatic fallback to secondary model tier.	Zero interruption to marketer UI; failure modes logged to BigQuery telemetry sink.
ADK Engine Fallback	Direct Agent Runtime RPC failure or schema serialization exception	Falls back to in-process deterministic orchestrator engine (app/orchestrator/engine.py).	100% availability guarantee; eliminates user-facing 500 internal server errors.

7. Deliverables Pipeline & Data Contract Architecture

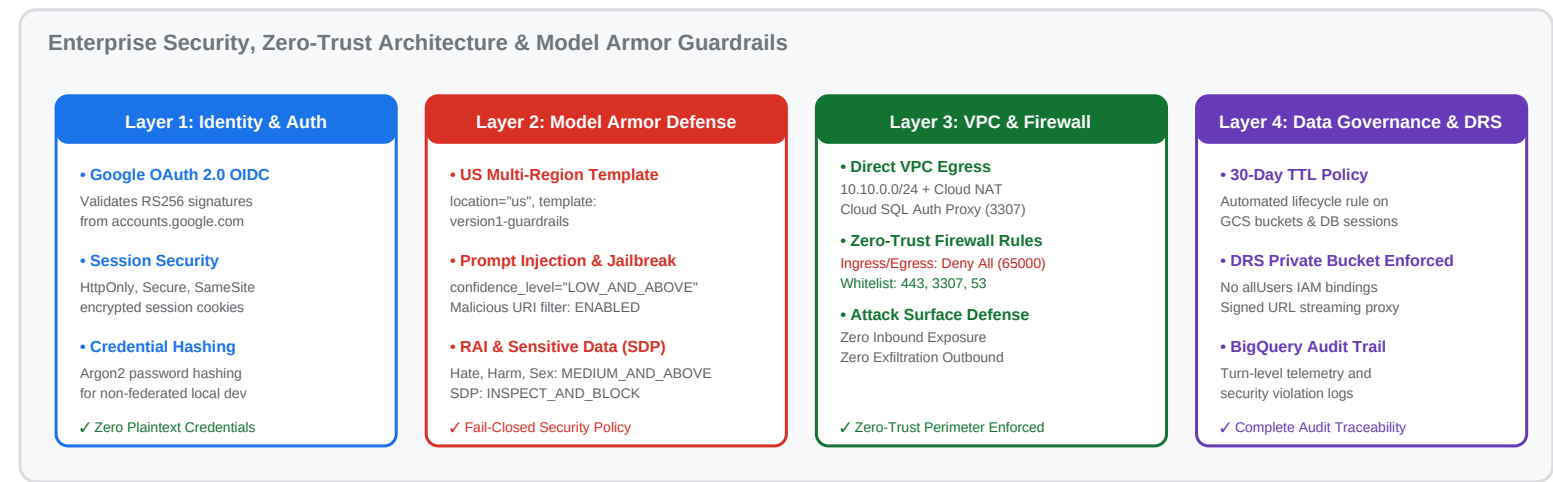
Contract-first data flow showing Pydantic v2 schemas passed between stages without context dilution.



Contract Schema	Pydantic v2 Model Path	Key Fields & Data Types	Validation & Error Handling
MarketSensingDeliverable	app.schemas.deliverables	keyTrends (list[str]), competitorInsights (list[dict]), sentimentScore (float)	Strict JSON mode enforcement; fallback to prompt objective on parse failure.
CampaignBriefDeliverable	app.schemas.deliverables	strategicNarrative (str), messagingPillars (list[str]), targetAudience (str)	Incorporates marketer revision feedback from stage history snapshots.
CreativeContentDeliverable	app.schemas.deliverables	adHeadlines (list[str]), visualPrompt (str), assetUrl (str)	Generates high-res image via Nano Banana 2 Lite (gemini-3.1-flash-lite-image); fallback to SVG placeholder.
PerformanceInsightsDeliverable	app.schemas.deliverables	expectedRoas (float), channelAllocations (list[ChannelAlloc]), predictedCpa	Programmatic normalization ensures channel percentages sum exactly to 100.0%.

8. Security, Zero-Trust Architecture & Model Armor Perimeter

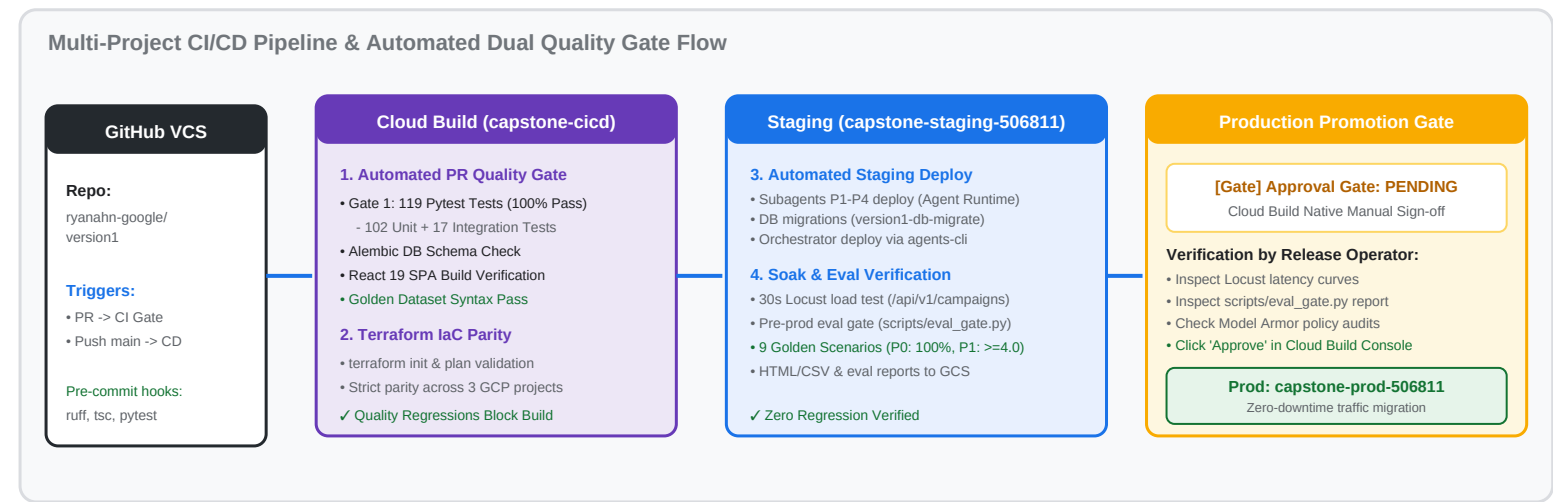
Four-tier defense-in-depth model encompassing identity, content guardrails, network isolation, and data governance.



Defense Layer	Enforced Mechanism	Threat Mitigated	Operational Policy & SLA
Content Defense	Google Cloud Model Armor (version1-guardrails in US Multi-Region)	Prompt injection, jailbreaks, system prompt leakage, toxic content.	Fail-closed policy (INSPECT_AND_BLOCK): blocks execution on security violation.
Identity & Auth	Google OAuth 2.0 OIDC + HttpOnly Secure Cookies	Session hijacking, CSRF attacks, unauthorized campaign manipulation.	Cryptographic token signature validation on every Cloud Run API call.
Network Isolation	Direct VPC Egress + Zero-Trust Firewall + Cloud SQL Auth Proxy	Inbound attack vectors, unauthorized outbound exfiltration, database credential interception.	Default-Deny Ingress & Egress (65000) ; Egress whitelisted strictly for TCP 443 (Google APIs), TCP 3307 (Cloud SQL Proxy mTLS), TCP/UDP 53 (DNS).
Data Governance & DRS	30-Day TTL Lifecycle Rule on GCS & DB + DRS compliance	Data hoarding, stale asset liability, storage cost creep, public link leakage.	100% private GCS bucket; signed URL proxy (1h validity); automated 30-day deletion.

9. Multi-Project CI/CD Pipeline & Automated Dual Quality Gate

Google Cloud Build pipeline enforcing 100% test passing, automated Golden Eval benchmark, and headless Locust load verification.



Pipeline Stage	Automated Action	Quality Gate Criterion	Promotion / Blocking Behavior
Pre-Commit Hook	Local developer git commit validation script	ruff format, ruff check, tsc --noEmit, pytest (119 tests)	Fails immediately if any check, linter, or test fails locally.
Pull Request (Gate 1)	Cloud Build trigger: pr-version1	119 pytest tests (102 unit + 17 int) + alembic check + npm build	100% pass required; merges to main are strictly blocked on failure.
Pre-Prod Eval (Gate 2)	scripts/eval_gate.py on 9 golden scenarios	P0: 100% Schema & Budget, 0 Failures; P1: Judge Score >= 4.0/5.0, Regression <= 0.2	Blocks release if prompt regression, schema drift, or budget math fails.
Staging Load Test	30s headless Locust verification against staging Cloud Run	0% HTTP error rate; P95 turn latency within SLA thresholds	Generates HTML/CSV audit reports; triggers production approval gate.
Production Promotion	Cloud Build trigger: deploy-version1	approval_config { approval_required = true }	Enters PENDING state until release operator manually approves in console.